

Don't Dwell on Opinion Stability

By Nicholas C. Dias

Abstract:

Since Converse's classic work, scholars have assumed that temporally unstable opinions are non-attitudes: shallow opinions that poorly reflect a citizen's core values or interests. But this assumption has not been carefully scrutinized. This article offers a broad test of whether "principled" opinions—i.e., opinions that *do* align with a citizen's values or interests—are more stable over time. Within high-quality panel datasets, I identify pairs of values and policy issues wherein the value entails an opinion about the issue. I then examine whether citizens are more likely to hold onto policy opinions that align with their values. I find that they consistently do—even after controlling for panelist, policy issue, value, and period fixed effects. However, the size of this effect is small, suggesting that temporal stability is a weak signal of how principled an opinion is. As such, scholars should rethink the implications of the vast literature on opinion stability.

WORKING PAPER – PLEASE DO NOT CIRCULATE

Can citizens' opinions meaningfully guide government policy? Many political scientists fear that the answer is no. While citizens often voice opinions about public policies, these "opinions" fluctuate considerably over time (Converse 1964; Freeder, Lenz, and Turney 2019; Kinder and Kalmoe 2017). This has led scholars to conclude that—for most citizens, on most policy issues—expressed opinions are "non-attitudes" that poorly reflect a citizen's underlying values or interests (Converse 1964; Zaller 1992). This problem threatens to short-circuit substantive representation: If citizens' policy opinions are a faint simulacrum of their values and interests, citizens may commonly vote for policy referendums and political candidates that will work against them (Ansolabehere, Rodden, and Snyder 2008; Mummolo, Peterson, and Westwood 2021). If this is the case, in Achen's (1975) words, "[d]emocratic theory loses its starting point" (1220).

Yet, judging whether a policy opinion is "principled"—i.e., consistent with a citizen's underlying values or interests—is challenging. Studies have often assumed that principled opinions do not change much over time whereas unprincipled opinions do (Converse 1964; Zaller 1992). But this assumption has not been carefully scrutinized, and it is intuitively questionable. After all, citizens can hold onto unprincipled opinions for myriad reasons—e.g., out of blind loyalty to their political party (Freeder, Lenz, and Turney 2019). And in many circumstances, principled opinions may be associated with opinion change—e.g., when citizens receive good information about policies (Bullock 2011; Jerit 2009).

This article offers a straightforward test of whether principled opinions are more stable over time, focusing on one important class of principles: citizens' values. Across two high-quality panel datasets, I identify pairs of values and policy issues wherein the value (e.g., compassion for others) logically entails a particular opinion about the issue (e.g., support for welfare). I then examine whether citizens are more likely to hold onto policy opinions that align with their values. I find that principled opinions are indeed more stable over time—even after controlling for panelist, policy issue, value, and period fixed effects. But the size of this effect is small: A two-standard-deviation increase in value-consistency corresponds to just a four-percentage-point increase in opinion stability.

These results suggest that temporal stability is a weak signal of how principled a policy opinion is. As such, scholars should rethink the implications of the vast literature on opinion stability. Namely, scholars cannot conclude that substantive representation is in jeopardy simply because citizens' policy opinions are unstable. Insofar as scholars want to judge whether citizens hold principled opinions—and thus constrain government policy—researchers must find other methods for evaluating citizens' opinions.

Why Worry About Opinion Stability?

Many scholars worry that not enough citizens hold principled views about policy issues. This is for good reason. Democracies are supposed to enact policies that reflect citizens' values and interests (Mansbridge 1983; Miller and Stokes 1963; Price and Neijens 1997). But realizing this ideal depends on whether citizens can translate their principles into concrete, actionable policy opinions. If citizens cannot apply their values and interests to the political world, they are ill-equipped to vote for political candidates and policy referendums that align with these principles (Ansolabehere, Rodden, and Snyder 2008; Mummolo, Peterson, and Westwood 2021). Moreover, elected representatives look to citizens' policy opinions to decide which policies to enact (Butler and Nickerson 2011; cf. Kalla and Porter 2021). If these opinions do not reflect citizens' principles, neither may government policy.

One key reason scholars doubt that many citizens hold principled opinions about most policy issues is that citizens' opinions are *unstable over time* (Converse 1964; Freeder, Lenz, and Turney 2019; Kinder and Kalmoe 2017). In his seminal article on belief systems, Converse (1964) argued that opinion stability “in itself is *prima facie* evidence that the belief has extremely low centrality for the believer,” (47). Unstable opinions are not derived from, or constrained by, “some superordinate value or posture toward man and society,” (ibid., 7). In other words, they are a poor reflection of a citizen's core values and interests—*unprincipled*. This argument remains influential today, justifying a decades-long debate over precisely how variable citizens' opinions are (Achen 1975; Ansolabehere, Rodden, and Snyder 2008; Converse 1964; Erikson 1979; Feldman 1989; Freeder, Lenz, and Turney 2019; Hill 2001; Kinder and Kalmoe 2017; Lauderdale, Hanretty, and Vivyan 2018).

Most studies on opinion stability are exercises in variance decomposition, whereby scholars attempt to calculate what proportion of the over-time variation in citizens' opinions reflects measurement error from vague policy questions, as opposed to genuine change resulting from citizens' ambivalence or ignorance about policy issues (Hill 2001; Feldman 1989; Lauderdale, Hanretty, and Vivyan 2018). This debate is still ongoing, with different methodologies producing different results (Ansolabehere, Rodden, and Snyder 2008; cf. Freeder, Lenz, and Turney 2019). Yet, few have questioned the underlying assumption that *opinion stability indicates how principled an opinion is*. This assumption is questionable, however.

Indeed, evidence from political science and psychology suggests that people hold onto unprincipled opinions for various reasons. The most obvious of these is *partisan motivated reasoning*: Citizens may adopt the opinions advocated by their political party (Bakker, Lelkes, and Malka 2020;

Barber and Pope 2019) and later resist changing these opinions, because doing so would imply that their political party is wrong (Bolsen, Druckman, and Cook 2014; Taber and Lodge 2006). Alternatively, citizens may selectively consume political news that favors their political party, such that their opinions are never seriously challenged (Kim and Kim 2021; Stroud 2010). In either case, citizens' opinions may hold onto unprincipled opinions simply because they are endorsed by their political party. Indeed, Freeder, Lenz, and Turney (2019) find that citizens primarily hold stable opinions when they know where their party stands on an issue and agree with their party. Yet, other plausible reasons that a citizen might hold onto an unprincipled opinion include—but are by no means limited to—a general distaste for uncertainty (Kruglanski and Webster 1996) or over-confidence in one's pre-existing opinions (Anson 2022).

There are also good reasons to think that, in many circumstances, principled opinions may change, rather than stand firm, as circumstances change. Most obviously, citizens *ought* to shift their policy opinions when presented with credible information that suggests a different opinion is more consistent with their principles. Indeed, they frequently do just that. In experiments and in the real world, citizens change their opinions when exposed to diagnostic information about policies (Bullock 2011; Gilens 2001; Fenger 2025; Jerit 2009; Thorson 2024). These changes appear to reflect that citizens update their policy opinions to become more consistent with their principles (Boudreau and MacKenzie 2018; Dias 2024). In fact, some evidence suggest that, when citizens' policy opinions do change in the real world, they tend to become *more* consistent with their principles (Vishwanath 2025).

One can also readily imagine situations where the same opinion, under different circumstances, is more or less consistent with a citizen's principles. Many Americans, for example, endorse the value of *deservingness*—rewarding people for their efforts and shielding them from the consequences of bad luck (Dias 2024). Among these individuals, we should expect support for welfare programs to increase following negative economic shocks, such as natural disasters or pandemics. That is, in order for these individuals to maintain principled opinions about welfare, they may very well need to change these opinions over time.

Finally, evidence for the idea that principled opinions are more stable is indirect. For instance, several scholars have pointed to the fact that politically sophisticated citizens—e.g., those high in education, political interest, or political knowledge—tend to have more stable opinions (Converse 2000; Converse and Pierce 1986; Dean and Moran 1977; Delli Carpini and Keeter 1996; Feldman 1989; Kinder and Kalmoe 2017). Certainly, one possible reason for this difference is that sophisticates are more likely to form principled opinions (Zaller 1992; cf. Goren, Smith, and Motta 2022). However,

it is well-documented that political sophisticates are also better at motivated reasoning (Bakker, Lelkes, and Malka 2020; Taber and Lodge 2006). As such, the relationship between political sophistication and opinion stability could reflect that sophisticates are better at defending their opinions, regardless of whether those opinions are principled.

Others have found relationships between opinion stability and supposed perceptual correlates of holding a principled opinion (Howe and Krosnick 2017). For example, Krosnick (1988; 1990) finds that citizens' opinions about personally *important* issues are more stable over time, which he attributes in part to "the linkage between these attitudes and core values," (Krosnick 1990, 68). On the other hand, using a wider range of panel datasets, Leeper (2014) finds that the effect of the importance of an opinion on its stability is small and statistically unreliable. In another study, Luttrell and Togans (2021) find that opinion stability is correlated with an opinion's *moral conviction*—i.e., whether a citizen perceives the opinion as grounded in their notions of right and wrong. However, psychological studies of the "introspection illusion" suggest that people are unreliable in identifying the causes of their opinions (for a review, see Pronin 2009). Indeed, some suggest that it is common for citizens' values to influence their policy opinions without citizens being aware of this influence (Jost 2021), and many citizens appear to misunderstand how their values should inform their policy opinions (Zaller 1992).

To determine whether principled opinions are more stable over time, one requires a direct measure of whether an opinion is principled, and ideally one that can vary within subjects. To the author's awareness, only one study has explicitly examined the relationship between citizens' principles and the stability of related opinions (Peffley and Hurwitz 1993). Yet, this study only examined changes in foreign policy attitudes over the course of a year, among residents of a small U.S. city, and so the generalizability of these findings is unclear. Moreover, the study relied on between-subjects comparisons between citizens with more or less principled opinions, leaving open the possibility of subject-level confounders.

The goal of this article is to provide a more definitive and generalizable answer to the question of whether principled opinions are more stable. Yet, before proceeding to my analysis, one point is worth acknowledging: A major appeal of using opinion stability as a proxy for opinion principledness is that it does not require researchers to stipulate and measure the principles that underlie citizens' opinions on a particular issue. Indeed, in many cases, scholars may not know all of the principles that could affect citizens' opinions about an issue. Yet, this does not mean that opinion stability cannot and should not be validated as a proxy for opinion principledness. One can still identify a subset of principles that are likely to be relevant to citizens' judgments about an issue. If these principles have a

negligible effect on opinion stability, one has little empirical reason to believe that stability is a good gauge of how principled an opinion is.

Study 1: The Effect of Human Values on Opinion Stability

To examine whether principled opinions are more stable, I require principles to compare policy opinions against. For my first study, I follow past scholarship by focusing on the human values of *self-transcendence* and *conservation* (Goren et al. 2016; Rathbun et al. 2016; Schwartz, Caprara, and Vecchione 2010; Schwartz et al. 2014). Broadly speaking, self-transcendence values encapsulate a person's concern for the welfare and interests of others, especially those who are different from them. Conservation values encapsulate a person's desire for social order, physical safety, and the preservation of cultural and religious customs (Schwartz 1992).

For the purposes of this study, self-transcendence and conservation values offer key advantages over other principles. First, unlike other principles, these values provide their endorsers with relatively clear guidance on which policies to support. Indeed, across many political cultures, self-transcendence values predict support for economic redistribution, racial equality, and immigration, as well as the prioritization of civil liberties over societal security (Goren et al. 2016; Rathbun et al. 2016; Schwartz, Caprara, and Vecchione 2010; Schwartz et al. 2014). By contrast, conservation values predict support for strict crime policies and military interventions abroad, as well as a prioritization of societal security over civil liberties (ibid.) Conservation values also predict opposition to changes in society's rules and/or composition: immigration, attempts to increase racial equality, and progressive social policies such as gay marriage (ibid.).

Second, self-transcendence and conservation values are broadly applicable to many policy issues. Indeed, across many political cultures, self-transcendence values predict support for economic redistribution, racial equality, and immigration, as well as the prioritization of civil liberties over societal security (Goren et al. 2016; Rathbun et al. 2016; Schwartz, Caprara, and Vecchione 2010; Schwartz et al. 2014). By contrast, conservation values predict support for strict crime policies and military interventions abroad, as well as a prioritization of societal security over civil liberties (ibid.). Conservation values also predict opposition to changes in society's rules and/or composition: immigration, attempts to increase racial equality, and progressive social policies such as gay marriage (ibid.). By contrast, other principles have more limited applicability. Take self-interest: whether a policy has a short- or medium-term impact on the material well-being of an individual or their immediate family (Sears and Funk 1991, 16). Self-interest has a substantial influence on citizens' policy opinions

in some circumstances (Doherty, Gerber, and Green 2006; Hall and Yoder 2022; Haselswerdt 2020; Horowitz and Levendusky 2011; Green and Gerken 1989). However, most citizens do not have a clear and substantial self-interest in the resolution of many policy issues (for a review, see Sears and Funk 1991).¹

Third, human values are “pre-political” and thus plausibly exogenous to citizens’ policy opinions. These values manifest across diverse cultures, suggesting that they are deeply rooted in evolutionary psychology (Schwartz 1992). Indeed, genetics explain a large proportion of the variation in these values (Cieciuch, Davidov, and Algesheimer 2016; Knafo and Spinath 2011; Schermer et al. 2008; Uzefovsky, Döring, and Knafo-Noam 2016; Vukasović and Bratko 2015). Human values drive behaviors outside of the political domain, from religiosity to career choice and even shopping behavior (Lee et al. 2022; Sagiv et al. 2017). People typically develop human values before they acquire policy opinions: Human values emerge by age eleven (Döring et al. 2015), but quasi-experimental evidence suggests that political socialization continues into late adolescence (Brown et al. 2023). Finally, human values are reasonably stable over long periods of time (Milfont, Milojev, and Sibley 2016; Vecchione et al. 2016; 2020) whereas policy opinions are not (Converse 1964; Kinder and Kalmoe 2017). As such, if value-consistency and opinion stability are related, that relationship is more likely to reflect an effect of value-consistency on opinion stability than vice versa.

Data and Analytic Approach

For my analysis, I rely on the German Longitudinal Election Study (GLES). This dataset includes measures of human values, and 24 policy opinions related to these values, from over 26,050 unique citizens across 15 waves (2016–2021). To my awareness, this is the *only* panel dataset that includes measures of human values and a large number of related policy opinions. The values of self-transcendence ($\alpha = .78$) and conservation ($\alpha = .69$) were measured by averaging items from the Portrait Values Questionnaire (Schwartz et al. 2001). These values were measured once for each panelist, at the time they joined the GLES. As such, I must use these initial measures to impute citizens’ values at subsequent waves, and I cannot independently verify that these values were stable over the course of the panel. However, past research finds that these values are reasonably stable over multi-year periods (Milfont, Milojev, and Sibley 2016; Vecchione et al. 2016; 2020), and I run robustness checks

¹ For example, consider policies like financial regulation, defense, legal and prison reforms, science and technology, immigration, agriculture, or foreign affairs.

to ensure that value change does not bias my results. The average amount of time between consecutive waves is approximately 169 days ($SD = 194.59$ days).

Figure 1. Restructuring the GLES Data

Panelist	Wave	SelfTrans	Conservation	OpinionB	OpinionC
A	1	0.6	0.3	0.6	0.5
A	2	0.6	0.3	0.7	0.5
A	3	0.6	0.3	0.6	0.5

Panelist	ValueID	ValueScore	PolicyIssue	CurrentOpinion	NextOpinion	CurrentWave	NextWave
A	Self-Trans.	0.6	B	0.6	0.7	1	2
A	Self-Trans.	0.6	B	0.7	0.6	2	3
A	Conservation	0.3	C	0.5	0.5	1	2
A	Conservation	0.3	C	0.5	0.5	2	3

Note: The GLES data were restructured such that each row represented a pair of consecutive policy-opinion measurements from one panelist. Observations from different periods and policy opinions were stacked on top of one another.

I rescale all measures of values and policy opinions so that they range from 0–1. Then, as depicted in Figure 1, I restructure the GLES data so that each row represents a particular panelist’s (i) attitudes toward a particular policy issue (p) and value (v) over a particular period (t), stacking observations from different panelists, values, policies, and periods on top of one another. I then model whether value-consistent opinions are more stable over time by running the following OLS regression:

$$NextOpinion_{i,p,t} = \beta_1 * CurrentOpinion_{i,p,t} + \beta_2 * Value-Consistency_{i,p,v,t} + \beta_3 * CurrentOpinion_{i,p,t} * Value-Consistency_{i,p,v,t} + \theta_i + \gamma_p + \psi_v + \lambda_t + \varepsilon_{i,p,v,t}$$

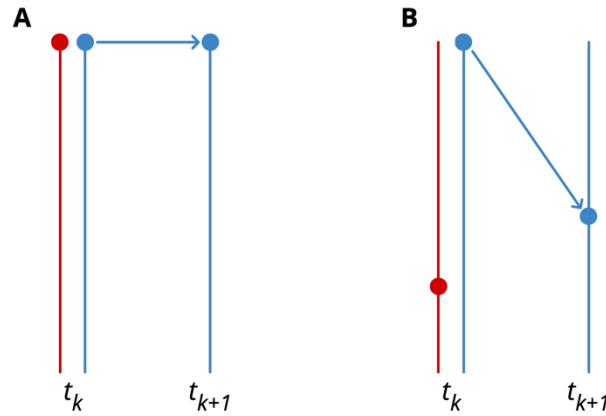
In this model, $NextOpinion_{i,p,t}$ represents panelist i ’s opinion about policy issue p at the end of time-period t , whereas $CurrentOpinion_{i,p,t}$ represents i ’s opinion about the same issue at the beginning of period t . $Value-Consistency_{i,p,v,t}$ represents how consistent $CurrentOpinion_{i,p,t}$ is with value v . This is measured by calculating the absolute distance between a citizen’s score on a value (e.g., self-transcendence) and their policy opinion.² Inverting this difference provides a measure of value-consistency: $Value-Consistency_{i,p,v,t} = 1 - |Opinion_{i,p,t} - Value_{i,v,t}|$. Finally, θ_i represents a vector of panelist fixed effects, γ_p

² Before performing this calculation, I “flip” the policy scales so that higher scores on the opinion indicate consistency with higher scores on the value. Note that a policy opinion can be related to both self-transcendence and conservation values.

represents a vector of policy-issue fixed effects, ψ_v represents a vector of value fixed effects, and λ_t represents a vector of period fixed effects. I cluster standard errors by panelist.

The structure of this model is somewhat complex. However, the comparisons that it makes can be stated intuitively. Imagine two time points. At t_k , we observe some value and a related policy opinion. At t_{k+1} , we observe opinions about the same policy issue again. In one scenario, depicted in Figure 2A, an individual's policy opinion (in blue) is highly consistent with a related value (in red). In this scenario, we would expect the individual's policy opinion to change little from t_k to t_{k+1} . In another scenario, depicted in Figure 2B, the individual's policy opinion is *less* consistent with a related value. In this scenario, an individual's policy opinion should change more from t_k to t_{k+1} .

Figure 2. Depicting the Model Comparison



Note: A stylistic depiction of the comparison being made by my model. The red dot-line represents a panelist's position on some value at t_k . The blue dot-lines represent the same panelist's position on a related policy opinion at t_k and t_{k+1} .

My model makes these same comparisons across multiple individuals, policy issues, values, and periods. In the model, β_1 quantifies the stability of a totally value-inconsistent opinion—i.e., an opinion that is maximally distant from some related value—over some period. β_3 quantifies whether stability *increases or decreases* as a panelist's opinion becomes more value-consistent. These coefficients can be interpreted similarly to an inter-wave correlation (β_1) and changes in such an inter-wave correlation (β_3).³ However, before estimating these coefficients, the model first disregards average differences between panelists, policy issues, values, and periods.

³ In contrast to some previous work (e.g., Converse 1964), I avoid measuring opinion stability with simple inter-wave correlations. This approach would be unwieldy, given the number of waves and policy items in the GLES. Nevertheless, my model yields a coefficients (e.g., β_1) that can be interpreted like inter-wave correlations. Indeed,

This modeling approach has several advantages. First, it allows me to estimate one coefficient (β_3) that represents a *general* tendency for value-consistent opinions to be more stable. Second, by construction, my dependent variable is always measured *after* my independent variables. This alleviates concerns about reverse causality. Third, this model leverages instances where the same citizen holds more or less value-consistent opinions on different policy issues within the same period, as well as instances where the same citizen holds more or less value-consistent opinions about the same issue at different periods. Thus, unlike previous studies, I need not rely on problematic between-subjects comparisons to estimate whether principled opinions are more stable over time. Fourth, the fixed effects in my model mean that the observed effects for value-consistency cannot be confounded by simple differences between panelists, policy issues, values, or periods. In particular, the period fixed effects account for differences in the length of time between panelists' consecutive waves and help to correct for potential non-stationarity in policy opinions.⁴

Results

Table 1 depicts the results of my model under different specifications. Across all specifications, value-consistent opinions are more stable than value-inconsistent opinions. For brevity, I quote the results from Column 4—the right-most model in Table 1 which includes panelist, issue, value, and period fixed effects.

When a citizen's policy opinion is totally *inconsistent* with some related value—i.e., when the absolute distance between the value and the policy opinion is at its maximum—the effect of a citizen's current policy opinion on their next policy opinion is roughly 0.59 ($SE = 0.003, p < .001$). This means that totally value-inconsistent opinions at t_k account for roughly 35 percent of the variation in opinions at t_{k+1} . As policy opinions become more value-consistent, they become slightly more stable ($b = 0.06, SE = 0.004, p < .001$). On average, the stability coefficient for an opinion that is totally *consistent* with a value is 0.65 ($SE = .002, p < .001$). In other words, totally value-consistent opinions at t_k account for roughly 42 percent of the variation in opinions at t_{k+1} . As I show in the online appendix, the

where the variance in opinions is equal at both waves, the inter-wave correlation is equal to the autoregressive coefficient. As shown in the online appendix, the means and variances of policy opinions do not exhibit a clear over-time trend (i.e., non-stationarity).

⁴ As shown in the online appendix, after residualizing by panelist, the time trends for policy opinions generally appear stationary. To further address concerns about stationarity, I conceptually replicate my analyses with a modeling approach that should be less vulnerable to stationarity violations and get qualitatively similar results to those I report here. In that approach, I model the relationship between value-consistency at the beginning of period t on the absolute change in policy opinions across period t .

interactive effect of value-consistency on opinion stability (β_3) is fairly consistent across different types of individuals and is consistently positive across policy issues.

Table 1. Opinions Aligned with Human Values are Slightly More Stable

	<u>Next Opinion</u>			
	(1)	(2)	(3)	(4)
Current Opinion	0.667*** (0.003)	0.658*** (0.003)	0.587*** (0.003)	0.587*** (0.003)
Value-Consistency of Current Opinion	-0.050*** (0.002)	-0.045*** (0.002)	-0.031*** (0.003)	-0.032*** (0.003)
Current Opinion * Value-Consistency of Current Opinion	0.067*** (0.003)	0.069*** (0.003)	0.059*** (0.004)	0.060*** (0.004)
(Intercept)	0.197*** (0.002)			
Num.Obs.	2,334,097	2,334,097	2,334,097	2,334,097
R2 Adj.	0.500	0.518	0.542	0.544
R2 Within Adj.		0.488	0.386	0.386
RMSE	0.21	0.21	0.21	0.21
FE: Panelist		X	X	X
FE: Value-Policy Pair			X	X
FE: Period				X

Note: Standard errors are clustered by panelist.

** $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$*

That said, the overall effect of value-consistency on opinion stability (β_3) is modest: Moving from the minimum to the maximum values of value-consistency increases opinion stability from 0.59 to just 0.65. And this constitutes a highly unrealistic amount of change in value-consistency: The standard deviation of value-consistency—after residualizing by panelist, issue, value, and period—is only 0.19. As such, a two-standard-deviation increase in value-consistency would increase opinion stability by about four percentage points. In short, I do not observe a strong relationship between opinion principledness and stability, as expected by existing research (e.g., Converse 1964).

Exploring Alternative Explanations

One may worry that the small effects of value-consistency on opinion stability are the result of bias in my measures of values. The GLES only measured each panelist's human values once, when they joined the GLES. As such, I must use these initial measures to impute citizens' values at subsequent waves. This choice could inject bias into my estimates of value-consistency's effects. For example, if values changed from t_k to t_{k+1} , value-consistency would be mismeasured at t_{k+1} , and I would misestimate the

effect of value-consistency at t_{k+1} on opinion stability from t_{k+1} to t_{k+2} . Insofar as values and policy opinions tend to shift in the same direction over time, as seems likely, the estimated effect of value-consistency at t_{k+1} would be biased downward.

However, past research strongly suggests that human values are stable over long periods, particularly among adults (Milfont, Milojev, and Sibley 2016; Vecchione et al. 2016; 2020). This suggests that any potential biases would be minor. Nonetheless, I run two robustness checks. First, if panelists' values shifted substantially over the course of the GLES, we should expect the effect of values (measured at each panelist's first wave) on policy opinions to decline as these measures become increasingly out-of-date. However, as I show in the online appendix, the average effect of values on policy opinions declines little over time (from $b = 0.28$ to $b = 0.23$ over the course of more than five years). Furthermore, I leverage the fact that individuals from 40–60 years old are the least likely to exhibit changes in their values (Milfont, Milojev, and Sibley 2016). As I show in the online appendix, when re-running my models on just those individuals who were between the ages of 40 and 55 when they joined the five-year panel, I see results that are qualitatively similar to those I report above.⁵

Alternatively, one may worry that the small effects of value-consistency on opinion stability are the result of error in my measures of policy opinions (e.g., Achen 1975). That is, measurement error may attenuate my estimates of the effects of value-consistency on opinion stability. To assess this possibility, I first gauge the amount of measurement error in my measures of policy opinions by running a simple, first-order autoregressive model for each policy issue:

$$NextOpinion_{i,p,t} = \beta_1 * CurrentOpinion_{i,p,t} + \gamma_p + \varepsilon_{i,p,t}$$

I find that citizens' opinions about different issues are on average relatively stable, judging by the autoregressive coefficient of β_1 ($Mean = 0.62$, $SD = 0.11$). These levels of stability would be surprising if the GLES policy items were tainted by significant measurement error.

Moreover, in the online appendix, I test the consequences of applying one approach to reducing measurement error: constructing average indexes from multiple policy items (Ansolabehere,

⁵ Of course, even if the GLES did measure values at every wave, it would remain possible for values to shift from t_k to t_{k+1} . This might complicate our interpretation of the effect of value-consistency at t_k on opinion stability from t_k to t_{k+1} . Yet, arguably, this possibility strengthens the argument of this article, in that it provides another reason to doubt that opinion stability is a good gauge for how principled an opinion is. By using opinion stability to gauge how principled an opinion is, one implicitly assumes that citizens' underlying values and interests are not shifting: If one's values change, then one should also shift their opinions in a similar direction, so as to maintain principled opinions. This is what Converse (1964) called *dynamic constraint*.

Rodden, and Snyder 2008). I find that measurement error tends to *inflate*, not attenuate, my estimates of the effects of value-consistency on opinion stability. This may be because the value-consistency measure helps to distinguish observations with more versus less measurement error. For example, value-inconsistent opinions may be more likely to reflect inattentive survey responding than value-consistent opinions.

Study 2: The Effect of Political Values on Opinion Stability

The findings of Study 1 cast doubt on the idea that principled opinions are considerably more stable over time. As such, the practice of using opinion stability to gauge whether citizens' opinions are principled may be suspect. But there may be another explanation for Study 1's findings, unaccounted for by my robustness checks. Despite evidence to the contrary, perhaps self-transcendence and conservation values are simply unimportant in shaping citizens' policy opinions (Goren et al. 2016; Rathbun et al. 2016; Schwartz, Caprara, and Vecchione 2010; Schwartz et al. 2014).

To address these concerns, I sought out panel datasets that contain pairs of *political* values and policy opinions that are so tightly related that we would strongly expect the value to have a large, stabilizing effect on the opinion. Take, for example, the value of *group equality* as measured by the SDO-Egalitarianism sub-scale (Ho et al. 2012; 2015) and support for attempts to make equal opportunities for ethnic minorities. In cases such as this, if I still find that value-consistency has a small effect on opinion stability, it would be difficult to imagine *any* principle having a large, stabilizing effect on a policy opinion.

Political values differ from human values in that they apply strictly to the political domain. Indeed, political values often apply even more narrowly to a subset of policy domains. For instance, *militarism* narrowly predicts opinions within the domain of foreign policy—e.g., defense spending and nuclear arms (Peffley and Hurwitz 1993). Thus, as compared to human values, the exogeneity of political values in relation to policy opinions is more questionable, as they lie in “close proximity” to policy opinions (Goren et al. 2016). However, several studies find that at least some political values are probably exogenous to policy opinions (Hatemi, Crabtree, and Smith 2019; Vishwanath 2025). We can leverage these values to provide another test of whether principled opinions are more stable over time.

Data and Analytic Approach

I rely on two large, high-quality panel datasets: the British Election Study (BES) Internet Panel and the German Longitudinal Election Study (GLES). **[AUTHOR’S NOTE: I PLAN TO ADD MORE PANEL DATASETS.]** Details about each panel are provided in Table 2. Within each dataset, I identify pairs of political values and policy issues wherein the value strongly implies a particular opinion about the issue. Pair examples are provided in Table 3, and all pairs are listed in the online appendix. To ensure that all political values were more stable than the policy opinions they were compared to, I limited myself to examining values (i) where I could verify that the value was more stable or (ii) that existing studies suggest are relatively stable over time (Evans and Neundorf 2020; Bratt, Sidanius, and Sheehy-Skeffington 2016; Milfont, Milojev, and Sibley 2016; Osborne et al. 2021; Vecchione et al. 2016; 2020).

Table 2. Panel Datasets Used in Study 2

Dataset	N	Num.Obs.	Waves	Period
British Election Study (BES) Internet Panel	39,244	751,703	29	Feb. 2014 – July 2024
German Longitudinal Election Study (GLES)	26,344	1,905,324	15	June 2016 – Dec. 2021

Table 3. Straightforward Pairings of Political Values and Policy Opinions

Political Value	Dataset	Policy Opinions
Group egalitarianism	BES	Support for attempts to increase equal opportunities for ethnic minorities, women, and homosexuals (respectively)
Environmentalism	GLES	Support for a carbon tax; prioritizing climate protection over economic growth
Nationalism	BES	Support for Brexit and ceding political power to the European Union

I maximize the number of observations in my analysis by filling in missing value observations with the last measurement of that value, to maintain consistency with Study 1. Finally, I restructure and analyze these data in the same manner as I did in Study 1, stacking observations from the different panel datasets on top of one another. I do not need to add a “country” fixed effect to the model, as political values and policy issues differ by country.

Results

Table 4 depicts the results across different model specifications. Again, across all specifications, value-consistent opinions are more stable. Moreover, the effects of value-consistency are considerably larger than they were in Study 1. Per Column 4 of Table 4, when a citizen's policy opinion is totally value-inconsistent, the effect of a citizen's current policy opinion on their next policy opinion is roughly 0.58 ($SE = 0.003$, $p < .001$). This means that totally value-inconsistent opinions at t_k account for roughly 34 percent of the variation in opinions at t_{k+1} . However, as policy opinions become more value-consistent, they become more stable ($b = 0.14$, $SE = 0.004$, $p < .001$). On average, the stability coefficient for a totally value-consistent opinion is 0.73 ($SE = .002$, $p < .001$). In other words, totally value-consistent opinions at t_k account for roughly 53 percent of the variation in opinions at t_{k+1} .

Table 4. Opinions Aligned with Political Values are Slightly More Stable

	<u>Next Opinion</u>			
	(1)	(2)	(3)	(4)
Current Opinion	0.667*** (0.002)	0.630*** (0.003)	0.581*** (0.003)	0.581*** (0.003)
Value-Consistency of Current Opinion	-0.097*** (0.002)	-0.085*** (0.002)	-0.091*** (0.002)	-0.092*** (0.002)
Current Opinion * Value-Consistency of Current Opinion	0.131*** (0.003)	0.142*** (0.004)	0.144*** (0.004)	0.144*** (0.004)
(Intercept)	0.193*** (0.001)			
Num.Obs.	2,657,027	2,657,027	2,657,027	2,657,027
R2 Adj.	0.554	0.593	0.607	0.609
R2 Within Adj.		0.514	0.441	0.442
RMSE	0.20	0.20	0.20	0.20
FE: Panelist		X	X	X
FE: Value-Policy Pair			X	X
FE: Period				X

Note: Standard errors are clustered by panelist.

** $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$*

Yet, again, moving from the bottom to the top of the value-consistency scale rarely happens in reality: The standard deviation of value-consistency—after residualizing by panelist, policy issue, value, and period—is only 0.20. Thus, a two-standard-deviation increase in value-consistency is associated with just a ten-percentage-point increase in opinion stability. There is simply no convincing evidence that principled policy opinions are substantially more stable over time.

Discussion

Past studies of the stability of citizens' policy opinions are some of the most thoughtful, rigorous, and well-cited within political science (Achen 1975; Ansolabehere, Rodden, and Snyder 2008; Converse 1964; Erikson 1979; Feldman 1989; Freeder, Lenz, and Turney 2019; Hill 2001; Kinder and Kalmoe 2017; Lauderdale, Hanretty, and Vivyan 2018). They are also well-motivated: If most citizens lack principled opinions about most policy issues, it is unclear whether citizens can direct their governments to pursue policies that serve their values and interests. However, this article suggests that these studies are built on a questionable assumption. Contrary to existing claims, principled opinions are only marginally more stable over time.

Bolstering this study's findings, there are several intuitive reasons to doubt that opinion stability is always indicative of a principled opinion. Various evidence suggests that citizens can hold onto their opinions for reasons that have little to do with whether a policy aligns with their principles. For instance, citizens may blindly adopt their party leaders' positions, which rarely change (Barber and Pope 2019; Freeder, Lenz, and Turney 2019). Moreover, citizens' opinions can also become more principled—e.g., when citizens are exposed to credible information about policies' consequences (e.g., Boudreau and MacKenzie 2018). Thus, the assumption that stable opinions are more principled is far from straightforward.

Even if principled opinions *were* substantially more stable, it is unclear that democracy requires citizens to hold principled opinions for long periods of time. This article has argued that policy opinions matter, at least in large part, because they constrain the policies that democracies enact (e.g., via votes for referendums and political candidates) and thus help to ensure these policies comport with citizens' values and interests. Yet, insofar as this is why they matter, citizens only need to hold principled opinions about the policies that are currently “on the ballot” in some way. Put simply, what is the democratic value of having a principled opinion about an issue such as deep-sea mining, if that issue is not on the political agenda?

Nonetheless, when there are policy decisions to be made, researchers ought to be concerned about whether citizens form principled opinions about these issues. How, then, should we judge whether citizens' opinions are principled? One approach may be to evaluate whether citizens have the skills to form principled policy opinions *when they need to*. As I describe elsewhere, this approach enjoys several advantages over existing approaches to evaluating citizens' opinions (see Chapter 4 of this dissertation). That is, this approach does not impose the unrealistic requirement that citizens hold policy opinions that are *perennially principled*.

Political scientists have long questioned whether citizens can meet the demands of democracy. In particular, a broad consensus suggests that citizens often fail to translate their values and interests into principled opinions about policy issues. This article has argued that—insofar as this consensus rests on claims about how much citizens’ opinions vary over time—it needs reconsideration. Over-time stability tells us very little about whether citizens hold principled opinions. If we wish to judge whether citizens hold principled opinions—as well we should, if we care about substantive representation—researchers must find better methods for evaluating citizens’ opinions.

References

- Achen, Christopher H. 1975. "Mass Political Attitudes and the Survey Response." *The American Political Science Review* 69 (4): 1218–31.
- Ansola-behere, Stephen, Jonathan Rodden, and James M. Snyder. 2008. "The Strength of Issues: Using Multiple Measures to Gauge Preference Stability, Ideological Constraint, and Issue Voting." *The American Political Science Review* 102 (2): 215–32.
- Anson, Ian G. 2022. "Epistemic Confidence Conditions the Effectiveness of Corrective Cues against Political Misperceptions." *Research & Politics* 9 (2): 205316802211078.
- Bakker, Bert N., Yphtach Lelkes, and Ariel Malka. 2020. "Understanding Partisan Cue Receptivity: Tests of Predictions from the Bounded Rationality and Expressive Utility Perspectives." *The Journal of Politics* 82 (3): 1061–77.
- Barber, Michael, and Jeremy C. Pope. 2019. "Does Party Trump Ideology? Disentangling Party and Ideology in America." *The American Political Science Review* 113 (1): 38–54.
- Bolsen, Toby, James N. Druckman, and Fay Lomax Cook. 2014. "The Influence of Partisan Motivated Reasoning on Public Opinion." *Political Behavior* 36 (2): 235–62.
- Boudreau, Cheryl, and Scott A. MacKenzie. 2018. "Wanting What Is Fair: How Party Cues and Information about Income Inequality Affect Public Support for Taxes." *The Journal of Politics* 80 (2): 367–81.
- Bratt, Christopher, Jim Sidanius, and Jennifer Sheehy-Skeffington. 2016. "Shaping the Development of Prejudice: Latent Growth Modeling of the Influence of Social Dominance Orientation on Outgroup Affect in Youth: Latent Growth Modeling of the Influence of Social Dominance Orientation on Outgroup Affect in Youth." *Personality & Social Psychology Bulletin* 42 (12): 1617–34.
- Brown, Jacob, Enrico Cantoni, Sahil Chinoy, Martin Koenen, and Vincent Pons. 2023. "The Effect of Childhood Environment on Political Behavior: Evidence from Young U.S. Movers, 1992–2021." w31759. Cambridge, MA: National Bureau of Economic Research. <https://doi.org/10.3386/w31759>.
- Bullock, John G. 2011. "Elite Influence on Public Opinion in an Informed Electorate." *The American Political Science Review* 105 (3): 496–515.
- Butler, Daniel M., and David W. Nickerson. 2011. "Can Learning Constituency Opinion Affect How Legislators Vote? Results from a Field Experiment." *Quarterly Journal of Political Science* 6 (1): 55–83.

- Cieciuch, Jan, Eldad Davidov, and René Algesheimer. 2016. "The Stability and Change of Value Structure and Priorities in Childhood: A Longitudinal Study: Values in Childhood." *Social Development (Oxford, England)* 25 (3): 503–27.
- Converse, Philip E. 1964. "The Nature of Belief Systems in Mass Publics." *Critical Review* 18 (1–3): 1–74.
- . 2000. "Assessing the Capacity of Mass Electorates." *Annual Review of Political Science (Palo Alto, Calif.)* 3 (1): 331–53.
- Converse, Philip E., and Roy Pierce. 1986. *Political Representation in France*. Cambridge, MA: Harvard University Press.
- Dean, Gillian, and Thomas Welling Moran. 1977. "Measuring Mass Political Attitudes: Change and Unreliability." *Political Methodology* 4 (4): 383–413.
- Delli Carpini, Michael X., and Scott Keeter. 1996. *What Americans Know about Politics and Why It Matters*. Yale University Press.
- Dias, Nicholas C. 2024. "Do Citizens' Policy Opinions Match Their Goals? Applying a New Test of Vertical Constraint."
- Doherty, Daniel, Alan S. Gerber, and Donald P. Green. 2006. "Personal Income and Attitudes toward Redistribution: A Study of Lottery Winners." *Political Psychology* 27 (3): 441–58.
- Döring, Anna K., Shalom H. Schwartz, Jan Cieciuch, Patrick J. F. Groenen, Valentina Glatzel, Justyna Harasimczuk, Nicole Janowicz, et al. 2015. "Cross-Cultural Evidence of Value Structures and Priorities in Childhood." *British Journal of Psychology (London, England: 1953)* 106 (4): 675–99.
- Erikson, Robert S. 1979. "The SRC Panel Data and Mass Political Attitudes." *British Journal of Political Science* 9 (1): 89–114.
- Evans, Geoffrey, and Anja Neundorff. 2020. "Core Political Values and the Long-Term Shaping of Partisanship." *British Journal of Political Science* 50 (4): 1263–81.
- Feldman, Stanley. 1989. "Measuring Issue Preferences: The Problem of Response Instability." *Political Analysis: An Annual Publication of the Methodology Section of the American Political Science Association* 1 (January): 25–60.
- Fenger, Jannik. 2025. "When Facts Matter: How Diagnostic Information Enlightens Public Opinion in Real-World Politics." Ph.D., Aarhus, Denmark: Aarhus University.
- Freder, Sean, Gabriel S. Lenz, and Shad Turney. 2019. "The Importance of Knowing 'What Goes with What': Reinterpreting the Evidence on Policy Attitude Stability." *The Journal of Politics* 81 (1): 274–90.

- Gilens, Martin. 2001. "Political Ignorance and Collective Policy Preferences." *The American Political Science Review* 95 (2): 379–96.
- Goren, Paul, H. Schoen, J. Reifler, T. Scotto, and W. Chittick. 2016. "A Unified Theory of Value-Based Reasoning and US Public Opinion." *Political Behavior*. <https://link.springer.com/article/10.1007/s11109-016-9344-x>.
- Goren, Paul, Brianna Smith, and Matthew Motta. 2022. "Human Values and Sophistication Interaction Theory." *Political Behavior* 44 (1): 49–73.
- Green, Donald Philip, and Ann Elizabeth Gerken. 1989. "Self-Interest and Public Opinion Toward Smoking Restrictions and Cigarette Taxes." *Public Opinion Quarterly* 53 (1): 1.
- Hall, Andrew B., and Jesse Yoder. 2022. "Does Homeownership Influence Political Behavior? Evidence from Administrative Data." *The Journal of Politics* 84 (1): 351–66.
- Haselswerdt, Jake. 2020. "Carving out: Isolating the True Effect of Self-Interest on Policy Attitudes." *The American Political Science Review* 114 (4): 1103–16.
- Hatemi, Peter K., Charles Crabtree, and Kevin B. Smith. 2019. "Ideology Justifies Morality: Political Beliefs Predict Moral Foundations." *American Journal of Political Science* 63 (4): 788–806.
- Hill, J. 2001. "An Extension and Test of Converse's 'Black-and-White' Model of Response Stability." *The American Political Science Review* 95 (2): 397–413.
- Ho, Arnold K., Jim Sidanius, Nour Kteily, Jennifer Sheehy-Skeffington, Felicia Pratto, Kristin E. Henkel, Rob Foels, and Andrew L. Stewart. 2015. "The Nature of Social Dominance Orientation: Theorizing and Measuring Preferences for Intergroup Inequality Using the New SDO₇ Scale." *Journal of Personality and Social Psychology* 109 (6): 1003–28.
- Ho, Arnold K., Jim Sidanius, Felicia Pratto, Shana Levin, Lotte Thomsen, Nour Kteily, and Jennifer Sheehy-Skeffington. 2012. "Social Dominance Orientation: Revisiting the Structure and Function of a Variable Predicting Social and Political Attitudes." *Personality & Social Psychology Bulletin* 38 (5): 583–606.
- Horowitz, Michael C., and Matthew S. Levendusky. 2011. "Drafting Support for War: Conscription and Mass Support for Warfare." *The Journal of Politics* 73 (2): 524–34.
- Howe, Lauren C., and Jon A. Krosnick. 2017. "Attitude Strength." *Annual Review of Psychology* 68 (1): 327–51.
- Jerit, Jennifer. 2009. "How Predictive Appeals Affect Policy Opinions." *American Journal of Political Science* 53 (2): 411–26.

- Jost, John T. 2021. *Left and Right: The Psychological Significance of a Political Distinction*. Oxford University Press.
- Kalla, Joshua L., and Ethan Porter. 2021. "Correcting Bias in Perceptions of Public Opinion among American Elected Officials: Results from Two Field Experiments." *British Journal of Political Science* 51 (4): 1792–1800.
- Kim, Jin Woo, and Eunji Kim. 2021. "Temporal Selective Exposure: How Partisans Choose When to Follow Politics." *Political Behavior* 43 (4): 1663–83.
- Kinder, Donald R., and Nathan P. Kalmoe. 2017. *Neither Liberal nor Conservative*. University of Chicago Press.
- Knafo, Ariel, and Frank M. Spinath. 2011. "Genetic and Environmental Influences on Girls' and Boys' Gender-Typed and Gender-Neutral Values." *Developmental Psychology* 47 (3): 726–31.
- Krosnick, Jon A. 1988. "Attitude Importance and Attitude Change." *Journal of Experimental Social Psychology*. <https://www.sciencedirect.com/science/article/pii/0022103188900388>.
- . 1990. "Government Policy and Citizen Passion: A Study of Issue Publics in Contemporary America." *Political Behavior* 12 (1): 59–92.
- Kruglanski, Arie W., and Donna M. Webster. 1996. "Motivated Closing of the Mind: 'Seizing' and 'Freezing.'" *Psychological Review* 103 (2): 263–83.
- Lauderdale, Benjamin E., Chris Hanretty, and Nick Vivyan. 2018. "Decomposing Public Opinion Variation into Ideology, Idiosyncrasy, and Instability." *The Journal of Politics* 80 (2): 707–12.
- Lee, Julie Anne, Anat Bardi, Paul Gerrans, Joanne Sneddon, Hester van Herk, Uwana Evers, and Shalom Schwartz. 2022. "Are Value–Behavior Relations Stronger than Previously Thought? It Depends on Value Importance." *European Journal of Personality* 36 (2): 133–48.
- Leeper, Thomas J. 2014. "Are Important Attitudes More Stable? No, Not Really."
- Luttrell, Andrew, and Lacount J. Togans. 2021. "The Stability of Moralized Attitudes over Time." *Personality & Social Psychology Bulletin* 47 (4): 551–64.
- Mansbridge, Jane J. 1983. *Beyond Adversary Democracy*. Chicago, IL: University of Chicago Press.
- Milfont, Tacio L., Petar Milojev, and Chris G. Sibley. 2016. "Values Stability and Change in Adulthood: A 3-Year Longitudinal Study of Rank-Order Stability and Mean-Level Differences: A 3-Year Longitudinal Study of Rank-Order Stability and Mean-Level Differences." *Personality & Social Psychology Bulletin* 42 (5): 572–88.
- Miller, Warren E., and Donald E. Stokes. 1963. "Constituency Influence in Congress*." *The American Political Science Review* 57 (1): 45–56.

- Mummolo, Jonathan, Erik Peterson, and Sean Westwood. 2021. "The Limits of Partisan Loyalty." *Political Behavior* 43 (3): 949–72.
- Osborne, Danny, Nicole Satherley, Todd D. Little, and Chris G. Sibley. 2021. "Authoritarianism and Social Dominance Predict Annual Increases in Generalized Prejudice." *Social Psychological and Personality Science* 12 (7): 1136–45.
- Peffley, Mark, and Jon Hurwitz. 1993. "Models of Attitude Constraint in Foreign Affairs." *Political Behavior* 15 (1): 61–90.
- Price, Vincent, and Peter Neijens. 1997. "OPINION QUALITY IN PUBLIC OPINION RESEARCH." *International Journal of Public Opinion Research* 9 (4): 336–60.
- Pronin, E. 2009. "Chapter 1 The Introspection Illusion." *Advances in Experimental Social Psychology* 41:1–67.
- Rathbun, Brian C., Joshua D. Kertzer, Jason Reifler, Paul Goren, and Thomas J. Scotto. 2016. "Taking Foreign Policy Personally: Personal Values and Foreign Policy Attitudes." *International Studies Quarterly: A Publication of the International Studies Association* 60 (1): 124–37.
- Sagiv, Lilach, Sonia Roccas, Jan Cieciuch, and Shalom H. Schwartz. 2017. "Personal Values in Human Life." *Nature Human Behaviour* 1 (9): 630–39.
- Schermer, Julie Aitken, N. T. Feather, Gu Zhu, and Nicholas G. Martin. 2008. "Phenotypic, Genetic, and Environmental Properties of the Portrait Values Questionnaire." *Twin Research and Human Genetics: The Official Journal of the International Society for Twin Studies* 11 (5): 531–37.
- Schwartz, Shalom H. 1992. "Universals in the Content and Structure of Values: Theoretical Advances and Empirical Tests in 20 Countries." In *Advances in Experimental Social Psychology*, edited by Mark P. Zanna, 25:1–65. Academic Press.
- Schwartz, Shalom H., Gian Vittorio Caprara, and Michele Vecchione. 2010. "Basic Personal Values, Core Political Values, and Voting: A Longitudinal Analysis." *Political Psychology* 31 (3): 421–52.
- Schwartz, Shalom H., Gian Vittorio Caprara, Michele Vecchione, Paul Bain, Gabriel Bianchi, Maria Giovanna Caprara, Jan Cieciuch, et al. 2014. "Basic Personal Values Underlie and Give Coherence to Political Values: A Cross National Study in 15 Countries." *Political Behavior* 36 (4): 899–930.
- Schwartz, Shalom H., Gila Melech, Arielle Lehmann, Steven Burgess, Mari Harris, and Vicki Owens. 2001. "Extending the Cross-Cultural Validity of the Theory of Basic Human Values with a Different Method of Measurement." *Journal of Cross-Cultural Psychology* 32 (5): 519–42.

- Sears, David O., and Carolyn L. Funk. 1991. "The Role of Self-Interest in Social and Political Attitudes." In *Advances in Experimental Social Psychology*, 1–91. Advances in Experimental Social Psychology. Elsevier.
- Stroud, N. 2010. "Polarization and Partisan Selective Exposure." *Journal of Communication* 60 (September):556–76.
- Taber, Charles S., and Milton Lodge. 2006. "Motivated Skepticism in the Evaluation of Political Beliefs." *American Journal of Political Science* 50 (3): 755–69.
- Thorson, Emily A. 2024. *The Invented State: Policy Misperceptions in the American Public*. Oxford University Press.
- Uzefovsky, Florina, Anna K. Döring, and Ariel Knafo-Noam. 2016. "Values in Middle Childhood: Social and Genetic Contributions: Social and Genetic Contributions to Children's Values." *Social Development (Oxford, England)* 25 (3): 482–502.
- Vecchione, Michele, Shalom Schwartz, Guido Alessandri, Anna K. Döring, Valeria Castellani, and Maria Giovanna Caprara. 2016. "Stability and Change of Basic Personal Values in Early Adulthood: An 8-Year Longitudinal Study." *Journal of Research in Personality* 63 (August):111–22.
- Vecchione, Michele, Shalom H. Schwartz, Eldad Davidov, Jan Cieciuch, Guido Alessandri, and Gilda Marsicano. 2020. "Stability and Change of Basic Personal Values in Early Adolescence: A 2-Year Longitudinal Study." *Journal of Personality* 88 (3): 447–63.
- Vishwanath, Arjun. 2025. "The Impact of Values on Issue Stances: Evidence from Panel Studies." *British Journal of Political Science* 55 (e50): e50.
- Vukasović, Tena, and Denis Bratko. 2015. "Heritability of Personality: A Meta-Analysis of Behavior Genetic Studies." *Psychological Bulletin* 141 (4): 769–85.
- Zaller, John. 1992. *The Nature and Origins of Mass Opinion*. Cambridge, England: Cambridge University Press.

Online Appendix for “Don’t Dwell on Opinion Stability”

Nicholas C. Dias

May 2, 2025

Contents

1	Question Wordings (Study 1)	1
2	Question Wordings (Study 2)	3
2.1	German Longitudinal Election Study	3
2.2	British Election Study (BES) Internet Panel	5
3	Stationarity of Policy Opinions	7
4	Heterogenous Treatment Effects (Study 1)	8
4.1	Panelist-Level Moderators	8
4.2	Issue-by-Issue Differences	8
5	Testing for Changing Values and Resulting Bias (Study 1)	10
6	Are Self-Transcendence / Conservation Values Just Unimportant?	11
7	Reducing Measurement Error with Policy Indexes	12

1 Question Wordings (Study 1)

Each row in the table below corresponds to the set of items used to measure a particular value (left side) and the policy opinions that were compared to that value in my analyses (right side).

Value	Related Policy Opinions
Self-Transcendence ($\alpha = .78$) • 3320c – He/she thinks it is important that every person in the world should be treated equally. He/she believes everyone should have equal opportunities in life. • 3320h – It is important to him/her to listen to people who are different from him/her. Even when he/she disagrees with them, he/she still wants to understand them. • 3320k – It is very important to him/her to help the people around him/her. He/she wants to care for their well-being. • 3320p – It is important to him/her to be loyal to his/her friends. He/she wants to devote himself/herself to people close to him/her. • 3320u – He/she strongly believes that people should care for nature. Looking after the environment is important to him/her.	• 1090 – Some people prefer lower taxes, although this results in less social services. Others prefer more social services, although this results in raising taxes. • 1130 – Should it be easier or more difficult for foreigners to immigrate? • 1210 – Some people think that foreigners should completely assimilate to the German culture. Others think that foreigners should be able to live according to their own culture. • 1411 – Some people think that the state should interfere without restrictions in the privacy and freedom of movement of citizens in order to combat terrorism. Others think that the privacy and freedom of movement of citizens should always be protected even if it hampers the fight against terrorism. • 2880b – Refugees who come to Germany for economic reasons should be deported. • 2880c – Islamic communities should be subject to surveillance by the state. • 2880d – The state should take measures to reduce differences in income levels. • 2880f – The exercise of the Islamic faith should be restricted in Germany. • 2880g – The state should stay out of the economy. • 2880h – State powers in the fight against crime should be extended, even though this will lead to increased surveillance of citizens. • 2880j – Rich citizens should pay more taxes in the future than they do now. • 2880l – Germany should provide financial support for EU member states experiencing great economic and financial difficulties. • 2880t – The state should save companies with many jobs from insolvency. • 2880x – Germany needs an annual upper limit ("Obergrenze") for refugees. • 2880ah – Germany should restrict the import of goods from other countries. • 2880ao – The European Union should do more to harmonise living conditions between EU countries. • 2880bb – In emergency situations, it is justified for the state to restrict civil liberties. • 2880bc – Health protection must take priority over promoting economic growth. • 4045a – Do you think that in the following respects the federal government has taken too few or too many measures to combat the corona crisis? Ensuring medical care

- 3320e – It is important to him/her to live in secure surroundings. He/she avoids anything that might endanger his/her safety.
- 3320j – It is important to him/her to be humble and modest. He/she tries not to draw attention to himself/herself.
- 3320m – It is important to him/her that the government ensures his/her safety against all threats. He/she wants the state to be strong so it can defend its citizens.
- 3320n – It is important to him/her always to behave properly. He/she wants to avoid doing anything people would say is wrong.
- 3320q – He/she thinks people should follow rules at all times. He/she believes that people should do what they're told.
- 3320t – It is important to the person to preserve the customs that he/she has learned. He/she thinks it is best to do things the traditional way.

- 1130 – Should it be easier or more difficult for foreigners to immigrate?
- 1210 – Some people think that foreigners should completely assimilate to the German culture. Others think that foreigners should be able to live according to their own culture.
- 1411 – Some people think that the state should interfere without restrictions in the privacy and freedom of movement of citizens in order to combat terrorism. Others think that the privacy and freedom of movement of citizens should always be protected even if it hampers the fight against terrorism.
- 1483a – The use of military force is never justified.
- 1483f – War is sometimes necessary to protect a country's interests.
- 2880a – Homosexual civil partnerships should also have the right to adopt children as well.
- 2880b – Refugees who come to Germany for economic reasons should be deported.
- 2880c – Islamic communities should be subject to surveillance by the state.
- 2880f – The exercise of the Islamic faith should be restricted in Germany.
- 2880h – State powers in the fight against crime should be extended, even though this will lead to increased surveillance of citizens.
- 2880x – Germany needs an annual upper limit ("Obergrenze") for refugees.
- 2880y – Germany's defence expenditure should be increased over the next few years.
- 2880al – The European Union should do more to protect its external borders.
- 2880bb – In emergency situations, it is justified for the state to restrict civil liberties.

2 Question Wordings (Study 2)

Each row in the table below corresponds to the set of items used to measure a particular value (left side) and the policy opinions that were compared to that value in my analyses (right side).

2.1 German Longitudinal Election Study

Value	Related Policy Opinions
3320c – He/she thinks it is important that every person in the world should be treated equally. He/she believes everyone should have equal opportunities in life.	• 2880a – Homosexual civil partnerships should also have the right to adopt children as well. • 2880c – Islamic communities should be subject to surveillance by the state. • 2880f – The exercise of the Islamic faith should be restricted in Germany.
2880d – The state should take measures to reduce differences in income levels.	• 1090 – Some people prefer lower taxes, although this results in less social services. Others prefer more social services, although this results in raising taxes. • 2880j – Rich citizens should pay more taxes in the future than they do now.
2880g – The state should stay out of the economy.	• 2880j – Rich citizens should pay more taxes in the future than they do now. • 2880t – The state should save companies with many jobs from insolvency. • 2880af – All EU member states should adopt the euro as a common currency.
3320e – It is important to him/her to live in secure surroundings. He/she avoids anything that might endanger his/her safety.	• 1130 – Should it be easier or more difficult for foreigners to immigrate? • 1411 – Some people think that the state should interfere without restrictions in the privacy and freedom of movement of citizens in order to combat terrorism. Others think that the privacy and freedom of movement of citizens should always be protected even if it hampers the fight against terrorism. • 2880h – State powers in the fight against crime should be extended, even though this will lead to increased surveillance of citizens.
3320m – It is important to him/her that the government ensures his/her safety against all threats. He/she wants the state to be strong so it can defend its citizens.	• 1130 – Should it be easier or more difficult for foreigners to immigrate? • 1411 – Some people think that the state should interfere without restrictions in the privacy and freedom of movement of citizens in order to combat terrorism. Others think that the privacy and freedom of movement of citizens should always be protected even if it hampers the fight against terrorism. • 2880h – State powers in the fight against crime should be extended, even though this will lead to increased surveillance of citizens.
3320k – It is very important to him/her to help the people around him/her. He/she wants to care for their well-being.	• 1090 – Some people prefer lower taxes, although this results in less social services. Others prefer more social services, although this results in raising taxes. • 2880l – Germany should provide financial support for EU member states experiencing great economic and financial difficulties.

3320t – It is important to the person to preserve the customs that he/she has learned. He/she thinks it is best to do things the traditional way.	• 1210 – Some people think that foreigners should completely assimilate to the German culture. Others think that foreigners should be able to live according to their own culture. • 2880a – Homosexual civil partnerships should also have the right to adopt children as well. • 2880f – The exercise of the Islamic faith should be restricted in Germany.
3320u – He/she strongly believes that people should care for nature. Looking after the environment is important to him/her.	• 1290 – Some say that the fight against climate change should definitely take precedence, even if it impairs economic growth. Others say that the economic growth should definitely take precedence, even if it impairs the fight against climate change. • 2880w – From 2030 onwards, no new cars with petrol or diesel engines should be registered in Germany. • 2880ak – A tax should be introduced on the emission of carbon dioxide (CO₂).

2.2 British Election Study (BES) Internet Panel

Value	Related Policy Opinions
Tolerance for Uncertainty ($\alpha = .72$) - tolUncertain1 – I hate not knowing what the future holds - tolUncertain2 – I strongly prefer to be certain about the outcome before making a decision - tolUncertain3 – I hate uncertainty	- asylumMore – Do you think that Britain should allow more or fewer of the following kinds of people to come and live in Britain? Asylum Seekers - euMore – Do you think that Britain should allow more or fewer of the following kinds of people to come and live in Britain? Workers from other EU countries - familiesMore – Do you think that Britain should allow more or fewer of the following kinds of people to come and live in Britain? Families of people who already live here - noneuMore – Do you think that Britain should allow more or fewer of the following kinds of people to come and live in Britain? Workers from outside the EU - studentsMore – Do you think that Britain should allow more or fewer of the following kinds of people to come and live in Britain? Foreign students - happyEULeave – How happy or how disappointed are you that the UK voted to leave the EU? - immigSelf – Some people think that the UK should allow many more immigrants to come to the UK to live and others think that the UK should allow many fewer immigrants. - radical – We need to fundamentally change the way society works in Britain
Nationalism ($\alpha = .70$) - ethno1 – Britain has a lot to learn from other countries in running its affairs - ethno2 – I would rather be a citizen of Britain than of any other country in the world - ethno3 – There are some things about Britain today that make me ashamed to be British - ethno4 – People in Britain are too ready to criticise their country - ethno5 – The world would be a better place if people from other countries were more like the British - ethno6 – I am often less proud of Britain than I would like to be	- cancelBrexit – Parliament should cancel Brexit without a second referendum - happyEULeave – How happy or how disappointed are you that the UK voted to leave the EU? - EUIntegrationSelf – Some people feel that Britain should do all it can to unite fully with the European Union. Other people feel that Britain should do all it can to protect its independence from the European Union. - euLawsLevel – Some laws are better made at the European level - euParlOverride – The British Parliament should be able to override all EU laws
Traditionalism ($\alpha = .66$) - incremental1 – Which statement comes closer to your own views – even if neither is exactly right? (1) To understand the world, people must free their minds from established traditions. (2) Established traditions provide the wisdom necessary to understand the world. - incremental2 – Which statement comes closer to your own views – even if neither is exactly right? (1) If ideas have been around for a long time, they probably need to be updated. (2) If ideas have been around for a long time, they probably have wisdom in them. - incremental3 – Which statement comes closer to your own views – even if neither is exactly right? (1) The world is always changing, so we should adjust our view of moral behavior to those changes. (2) Even though the world is always changing, we should not adjust our view of moral behavior to those changes. - incremental4 – Which statement comes closer to your own views – even if neither is exactly right? (1) It is better to stick with established ideas that have proven their principle. (2) It is better to search for new ways of doing things that might be better.	- radical – We need to fundamentally change the way society works in Britain - femaleEquality – Attempts to give equal opportunities to women - gayEquality – Attempts to give equal opportunities to gays and lesbians - sovereignty1 – People in Scotland should have the ultimate right to decide for themselves how they should be governed

Group Egalitarianism ($\alpha = .75$)	
• sdoantieg1 – Group equality should not be our primary goal • sdoantieg2 – It is unjust to try to make groups equal • sdoantieg3 – We should do what we can to equalize conditions for different groups • sdoantieg4 – We should work to give all groups an equal chance to succeed	• blackEquality – Attempts to give equal opportunities to ethnic minorities • femaleEquality – Attempts to give equal opportunities to women • gayEquality – Attempts to give equal opportunities to gays and lesbians

3 Stationarity of Policy Opinions

Traditional autoregressive models depend on the assumption of *stationarity*—i.e., that the statistical properties of time-varying variables (e.g., mean, standard deviation) do not change over time. Figure 1 plots the average (Panel A) and standard deviation (Panel B) of each policy item by wave, after residualizing by panelist. As shown in Figure 1, the time trends for policy opinions generally appear stationary.



Figure 1: Panel A depicts the average of each policy item by wave, after residualizing by panelist. Panel B depicts the standard deviation of each policy item by wave, after residualizing by panelist.

To further address concerns about stationarity, I conceptually replicate my analyses with a “differencing” approach that should be less vulnerable to stationarity violations. That is, I regress the absolute amount of change in policy opinions on value-consistency:

$$|NextOpinion_{i,p,t} - CurrentOpinion_{i,p,t}| = \beta_1 * ValueConsistency_{i,p,v,t} + \theta_i + \gamma_p + \psi_v + \lambda_t + \epsilon_{i,p,v,t}$$

The results of these models, shown in Table 1, are qualitatively similar to those I report in the main paper. Looking at Column 1, we see that totally value-inconsistent opinions vary by an average of roughly 0.17 between waves, which is equivalent to 17 percent of the policy-opinion scale. Totally value-consistent opinions vary only slightly less, by an average of 0.14 between waves.

Table 1: Value-Consistent Opinions are Slightly More Stable

	(1)	(2)	(3)	(4)
Value-Consistency of Current Opinion	−0.030*** (0.001)	−0.011*** (0.001)	−0.026*** (0.001)	−0.026*** (0.001)
(Intercept)	0.168*** (0.001)			
Num.Obs.	2 334 097	2 334 097	2 334 097	2 334 097
R2 Adj.	0.001	0.088	0.103	0.106
R2 Within Adj.		0.000	0.001	0.001
RMSE	0.18	0.17	0.17	0.17
FE: Panelist		X	X	X
FE: Value-Policy Pair			X	X
FE: Period				X

Note: Standard errors are clustered by participant.

4 Heterogenous Treatment Effects (Study 1)

4.1 Panelist-Level Moderators

To examine whether the effects of value-consistency on opinion stability varied across different types of individuals, I tercile-split my sample across several moderators and re-ran my primary model separately on each tercile. Table 2 displays the predicted effect of *CurrentOpinion* on *NextOpinion*—i.e., opinion stability—at the minimum and maximum of *ValueConsistency*.

Table 2: Heterogenous Treatment Effects (Study 1)

Moderator	Tercile	$b_1 ValueConsistency = 0$	$b_1 ValueConsistency = 1$
Political Interest	Bottom	0.61 (0.005)	0.7 (0.003)
Political Interest	Middle	0.6 (0.005)	0.63 (0.003)
Political Interest	Top	0.55 (0.005)	0.58 (0.004)
Age	Bottom	0.58 (0.006)	0.66 (0.005)
Age	Middle	0.6 (0.006)	0.66 (0.005)
Age	Top	0.58 (0.007)	0.63 (0.005)
Education	Bottom	0.54 (0.005)	0.6 (0.004)
Education	Middle	0.58 (0.005)	0.63 (0.004)
Education	Top	0.63 (0.005)	0.69 (0.004)
Income	Bottom	0.57 (0.005)	0.64 (0.004)
Income	Middle	0.58 (0.005)	0.63 (0.004)
Income	Top	0.61 (0.005)	0.66 (0.004)

4.2 Issue-by-Issue Differences

To examine whether the effects of value-consistency varied by value and issue, I ran the following model separately on each pairing of value and policy issue:

$$NextOpinion_{i,t} = \beta_0 + \beta_1 * CurrentOpinion_{i,t} + \beta_2 * ValueConsistency_{i,t} + \beta_3 * CurrentOpinion_{i,t} * ValueConsistency_{i,t} + \lambda_t + \epsilon_{i,t}$$

In Figure 2, I plot the b_1 against the b_3 for each of the models. Point color indicates whether the value in the model was self-transcendence (blue) or conservation (red).

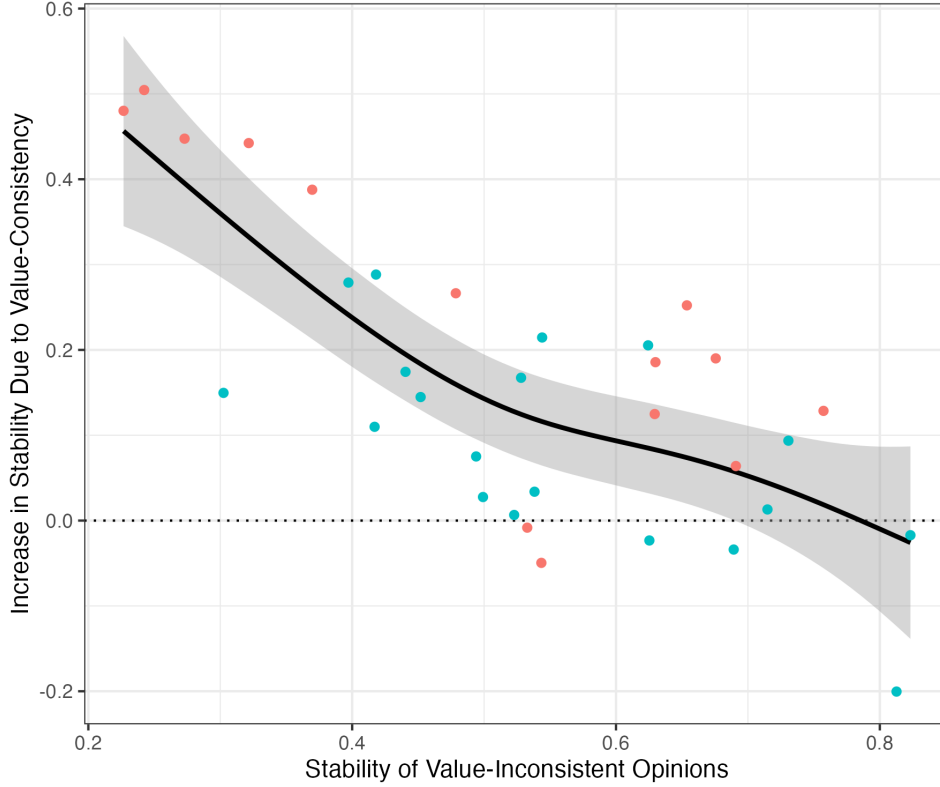


Figure 2: Distribution of β_3 's from issue-level models

Two patterns are evident from Figure 2. First, the effect of value-consistency on opinion stability is generally positive. Second, there is considerable variation depending on the pairing of value and policy issue and, in particular, the effect of value-consistency on opinion stability (b_3) decreases as the baseline stability of value-inconsistent opinions (b_1) increases. This variation does not appear to be due to policy subject matter: The same policy issues appear at both the top-left and bottom-right of the plot. Instead, it seems that some values may be more important for stabilizing certain opinions than others. For example, consider question 1411 from the GLES, which reads as follows:

Some people think that the state should interfere without restrictions in the privacy and freedom of movement of citizens in order to combat terrorism. Others think that the privacy and freedom of movement of citizens should always be protected even if it hampers the fight against terrorism. What is your opinion on this issue? (in favour of strong state interference - 1 / 2 / 3 / 4 / 5 / 6 / 7 - against strong state interference)

When citizens' responses to this question are totally inconsistent (versus consistent) with their *conservation* values, the autoregressive coefficient equals 0.23 (versus 0.71). By contrast, when citizens' responses to this question are totally inconsistent (versus consistent) with their *self-transcendence* values, the autoregressive coefficient equals 0.52 (versus 0.53). In this case, it appears that conservation values were more important for stabilizing citizens' opinions than self-transcendence values.

5 Testing for Changing Values and Resulting Bias (Study 1)

In Study 1, self-transcendence and conservation values were measured once per panelist, at the time they joined the GLES. Though previous research strongly suggests that these values are reasonably stable over long periods (Milfont et al. 2016; Vecchione et al. 2016; Vecchione et al. 2020), I cannot independently verify that self-transcendence and conservation values were stable over the course of the panel. However, I can test an implication of these values changing: If panelists' values shifted substantially over the course of the GLES, we should expect the effect of values (measured at each participant's first wave) on policy opinions to weaken as these measures become increasingly out-of-date. I ran the following model:

$$NextOpinion_{i,p} = \beta_1 * Value_{i,v} + \beta_2 * YearsSinceValueWasMeasured_{i,v} + \beta_3 * Value_{i,v} * YearsSinceValueWasMeasured_{i,v} + \theta_i + \gamma_p + \psi_v + \epsilon_{i,p,v}$$

As shown in Table 3, the average effect of values on policy opinions declines little over the course of a year ($b = -0.01$, $SE = 0.001$, $p < .001$). To put this in perspective, moving from the minimum to the maximum value of *Years Since Value Was Measured*—a difference of over five years—reduces the effect of *Value* from 0.28 to 0.23. In other words, 82 percent of the effect of values persists after five years.

Table 3: Effects of Values Over Time

	(1)
Value	0.281*** (0.004)
Years Since Value Was Measured	0.004*** (0.000)
Value * Years Since Value Was Measured	−0.010*** (0.001)
Num.Obs.	2 306 194
R2 Adj.	0.268
R2 Within Adj.	0.026
RMSE	0.27

Note: This model controls for panelist, policy issue, and value fixed effects. Standard errors are clustered by participant.

Another way to test whether changes in values over time are biasing my estimates is to subset to panelists whose values are unlikely to have changed over the course of the GLES. According to past research, individuals from 40–60 years old are the least likely to exhibit changes in their values (Milfont et al. 2016). As shown in Table 4, when re-running my models on just those individuals who were between 40 and 55 when they joined the five-year panel, I see results that are qualitatively similar to those I report in the main paper.

Table 4: Among the Middle-Aged, Principled Opinions Are Slightly More Stable

	(1)
Current Opinion	0.597*** (0.006)
Value-Consistency of Current Opinion	−0.031*** (0.006)
Current Opinion * Value-Consistency of Current Opinion	0.063*** (0.008)
Num.Obs.	746 601
R2 Adj.	0.572
R2 Within Adj.	0.402
RMSE	0.20

Note: This model controls for panelist, policy issue, value, and period fixed effects. Standard errors are clustered by participant.

6 Are Self-Transcendence / Conservation Values Just Unimportant?

Existing research suggests that self-transcendence and conservation values are an important source of structure in citizens' policy opinions, consistently predicting citizens' opinions across several policy domains and political cultures (Goren et al. 2016; Rathbun et al. 2016; Schwartz et al. 2010; Schwartz et al. 2014). Nonetheless, readers might worry that value-consistency has a modest effect on opinion stability because these values are unimportant. To address this concern, I calculate a common proxy for value importance: the extent to which a person's agreement (disagreement) with a value is more extreme than their agreement (disagreement) with other values (Schwartz et al. 2001). To do this, I first calculate each participant's average level of agreement with *all* human values measured by the GLES, not just self-transcendence and conservation values. Then, I subtract this cross-value average from the panelist's ratings of self-transcendence and conservation values, respectively, and take the absolute value:

$$\begin{aligned} &|SelfTranscendence_i - CrossValueAverage_i| \\ &|Conservation_i - CrossValueAverage_i| \end{aligned}$$

To estimate the moderating effect of value importance, I tercile-split my sample by value importance and re-run my primary model separately on each tercile. The results are displayed in Table 5. Indeed, values have a stronger stabilizing effect on policy opinions when they are important. However, even among those who most care about a value, the stabilizing effect of value-consistency is modest.

Table 5: Effect of Value-Consistency on Opinion Stability, by Value Importance

	(1)	(2)	(3)
Current Opinion	0.610*** (0.006)	0.639*** (0.005)	0.660*** (0.004)
Value-Consistency of Current Opinion	0.020*** (0.005)	−0.005 (0.004)	−0.038*** (0.004)
Current Opinion * Value-Consistency of Current Opinion	−0.029*** (0.008)	0.008 (0.007)	0.038*** (0.006)
Num.Obs.	771 229	779 228	783 640
R2 Adj.	0.525	0.569	0.586
R2 Within Adj.	0.348	0.411	0.465
RMSE	0.21	0.20	0.20

Note: This model controls for panelist, policy issue, value, and period fixed effects. Standard errors are clustered by participant.

7 Reducing Measurement Error with Policy Indexes

One may worry that the small effects of value-consistency on opinion stability are the result of error in my measures of policy opinions. That is, measurement error may attenuate my estimates of the effects of value-consistency on opinion stability. To address this concern, I test the consequences of applying one approach to reducing measurement error: constructing average indexes from multiple policy items.

First, in the GLES data, I searched for pairs of policy opinions that are similar conceptually and correlated empirically ($|r| \geq .3$). This yielded seven indexes capturing five distinct policy areas. Second, I ran the following model separately for each policy index and related value:

$$NextOpinion_{i,t} = \beta_1 * CurrentOpinion_{i,t} + \beta_2 * ValueConsistency_{i,t} + \beta_3 * CurrentOpinion_{i,t} * ValueConsistency_{i,t} + \epsilon_{i,t}$$

To provide a point of comparison, I re-ran this model after swapping the policy index with each constitutive policy item. For example, if I constructed an index from items A and B, I ran the above model using the index of items A and B, item A alone, and item B alone. Comparing the β_3 's across these models should provide a sense of how measurement error affects my estimates of the effect of value-consistency on opinion stability.

Figure 3 plots the average β_3 from the single item models (represented by points) against the corresponding β_3 from the policy-index model (represented by triangles). Line color indicates whether value-consistency was defined relative to self-transcendence (blue) or conservation (red) values. As shown in Figure 3, measurement error appears to *inflate* estimates of β_3 more often than it attenuates estimates of β_3 . This may be because the value-consistency measure helps to distinguish observations with more versus less measurement error. For example, value-inconsistent opinions may be more likely to reflect inattentive survey responding than value-consistent opinions.

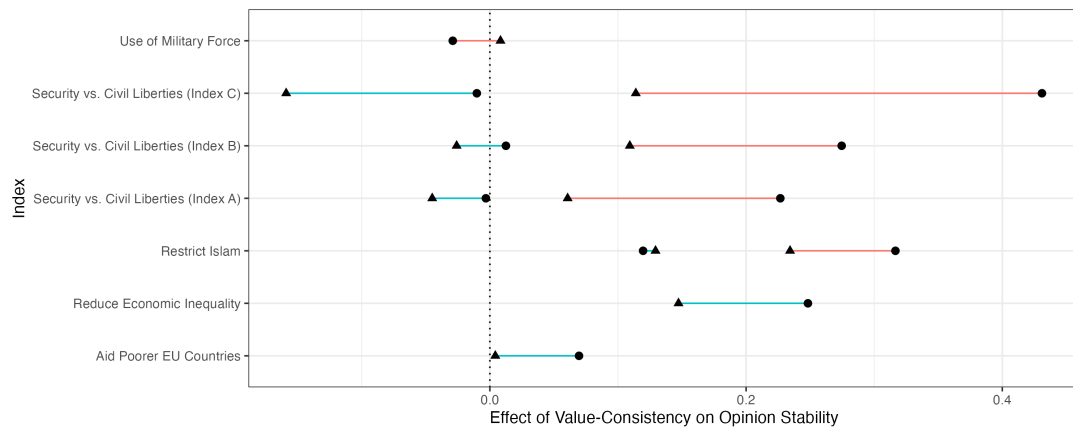


Figure 3: Difference in estimates of the effect of value-consistency on opinion stability (β_3) by use of policy indexes or corresponding policy items. Triangles represent the the β_3 from the policy-index model. Points represent the average β_3 from the single item models. Line color indicates whether value-consistency was defined relative to self-transcendence (blue) or conservation (red) values.